

MODUL PRAKTIK KEAMANAN SIBER

Brute Force Attack Simulation & Incident Response

Defense in Depth Lab Project — Tahap Lanjutan

Nama	Glenvio Regalito Rahardjo
Sekolah	SMK Telkom Purwokerto
Jurusan	TJKT — Cybersecurity Specialization
Environment	LAB-NET 192.168.10.0/24
Tanggal	17/02/2026

1. Project Overview & Objectives

Modul ini merupakan kelanjutan langsung dari project Defense in Depth (UFW + SSH Hardening + Fail2ban). Tujuannya adalah membuktikan bahwa sistem yang telah di-hardening mampu mendeteksi, menahan, dan merespons serangan brute force SSH secara otomatis.

Objectives

- **Simulate:** Mensimulasikan serangan brute force SSH menggunakan Hydra dari attacker node
- **Detect:** Membuktikan deteksi real-time melalui auth.log dan fail2ban.log
- **Contain:** Memverifikasi Fail2ban otomatis memblokir IP attacker
- **Respond:** Mendokumentasikan incident response sesuai framework NIST SP 800-61
- **Recover:** Memastikan sistem kembali normal setelah insiden

Framework yang Digunakan

NIST SP 800-61 — Computer Security Incident Handling Guide
Phase 1 — Preparation : Lab environment sudah terkonfigurasi dari modul DiD sebelumnya
Phase 2 — Detection : Monitoring auth.log + fail2ban.log secara real-time
Phase 3 — Containment : Fail2ban auto-block + manual verification
Phase 4 — Eradication : Verifikasi tidak ada sesi aktif yang berhasil masuk
Phase 5 — Recovery : Unban IP + validasi akses legitimate
Phase 6 — Lessons Learned : Dokumentasi dan rekomendasi peningkatan

Lab Environment

Host	IP Address	Role	OS
Ubuntu Server	192.168.10.100	🛡️ Defender	Ubuntu Server 24.04 LTS
Ubuntu Desktop	192.168.10.101	🔴 ⚔️ Attacker	Ubuntu Desktop 24.04

2. Phase 1 — Preparation

🎯 Tujuan Phase Ini

Memastikan semua komponen lab aktif dan siap sebelum simulasi serangan dimulai.

Catat semua output perintah berikut sebagai bukti baseline (kondisi SEBELUM serangan).

2.1 Verifikasi Status Layanan di Ubuntu Server

Jalankan semua perintah berikut di terminal Ubuntu Server (192.168.10.100):

Step 1 — Cek Status UFW Firewall

```
# Jalankan di: Ubuntu Server
sudo ufw status verbose

# Expected output:
# Status: active
# To          Action      From
# --          -
# OpenSSH     ALLOW IN    Anywhere
# 80/tcp      ALLOW IN    Anywhere
```

```
ubuntu@UbuntuServerPFFirewall:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To          Action      From
--          -
22/tcp (OpenSSH)    ALLOW IN    Anywhere
80/tcp          ALLOW IN    Anywhere
22/tcp (OpenSSH (v6))  ALLOW IN    Anywhere (v6)
80/tcp (v6)        ALLOW IN    Anywhere (v6)

ubuntu@UbuntuServerPFFirewall:~$
```

Step 2 — Cek Status Fail2ban

```
# Jalankan di: Ubuntu Server
sudo systemctl status fail2ban

# Expected: Active: active (running)
```

```
ubuntu@UbuntuServerPFFirewall:~$ sudo systemctl status fail2ban.service
• fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-02-17 08:16:21 UTC; 1min 42s ago
     Docs: man:fail2ban(1)
   Main PID: 570 (fail2ban-server)
      Tasks: 5 (limit: 4606)
    Memory: 40.7M (peak: 40.9M)
       CPU: 1.244s
   CGroup: /system.slice/fail2ban.service
           └─570 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Feb 17 08:16:21 UbuntuServerPFFirewall systemd[1]: Started fail2ban.service - Fail2Ban Service.
Feb 17 08:16:24 UbuntuServerPFFirewall fail2ban-server[570]: 2026-02-17 08:16:24,569 fail2ban.configreader
Feb 17 08:16:25 UbuntuServerPFFirewall fail2ban-server[570]: Server ready
lines 1-14/14 (END)
```

Step 3 — Cek Jail SSH (Baseline = 0 banned)

```
# Jalankan di: Ubuntu Server
sudo fail2ban-client status sshd
```

```
# Expected output:
# Status for the jail: sshd
# |- Filter
# |   |- Currently failed: 0
# |   `-- Total failed:    0
# `-- Actions
#     |- Currently banned: 0
#     `-- Banned IP list:
```

```
ubuntu@UbuntuServerPFFirewall:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:    0
|   `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd
`-- Actions
    |- Currently banned: 0
    |- Total banned:    0
    `-- Banned IP list:
```

2.2 Install Hydra di Ubuntu Desktop (Attacker)

Pindah ke terminal Ubuntu Desktop (192.168.10.101):

```
# Jalankan di: Ubuntu Desktop (Attacker)
sudo apt update && sudo apt install hydra -y
```

```
# Verifikasi instalasi:
hydra -h | head -5
```

```
ubuntu@UbuntuDesktopPFFIREWALL:~$ hydra -h | head -5
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations,
or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Syntax: hydra [[[-l LOGIN][-L FILE]] [-p PASS][-P FILE]] | [-C FILE]] [-e nsr] [-o FILE] [-t TASKS] [-M FILE [-T TASKS]] [-w TIME] [-W TIME] [-f] [-s PORT] [-x MIN:MAX:CHARSET] [-c TIME] [-ISOuvVd46] [-m MODULE_OPT] [service://server[:PORT][/OPT]]
```

```
# Cek wordlist rockyou.txt tersedia:  
ls -lh /usr/share/wordlists/rockyou.txt
```

```
ubuntu@UbuntuDesktopPFFIREWALL:~$ ls -lh /usr/share/wordlist/rockyou.txt  
-rw-rw-r-- 1 ubuntu ubuntu 134M Dec  8 2021 /usr/share/wordlist/rockyou.txt  
ubuntu@UbuntuDesktopPFFIREWALL:~$
```

2.3 Buka Monitoring Log di Ubuntu Server

Buka terminal BARU di Ubuntu Server (jangan tutup terminal sebelumnya) untuk monitoring real-time:

```
# Jalankan di: Ubuntu Server — Terminal TERPISAH  
sudo tail -f /var/log/auth.log  
  
# Biarkan terminal ini terbuka selama simulasi!  
# Kamu akan melihat aktivitas login muncul real-time di sini.
```

⚠ Penting

Gunakan 2 terminal di Ubuntu Server secara bersamaan:

Terminal A : Untuk menjalankan perintah administrasi

Terminal B : Untuk monitoring tail -f /var/log/auth.log (biarkan terus berjalan)

Kalian bisa membuka 2 terminal dengan menekan CTRL+ALT+F2, untuk Kembali bisa CTRL+ALT+F1.

Terminal Log :

```
ubuntu@UbuntuServerPFFirewall:~$ sudo tail -f /var/log/auth.log  
2026-02-17T08:18:56.027020+00:00 UbuntuServerPFFirewall sudo: ubuntu : TTY=ttty ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/fail2ban-client status sshd  
2026-02-17T08:18:56.027973+00:00 UbuntuServerPFFirewall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)  
2026-02-17T08:18:56.126122+00:00 UbuntuServerPFFirewall sudo: pam_unix(sudo:session): session closed for user root  
2026-02-17T08:25:01.845919+00:00 UbuntuServerPFFirewall CRON[1210]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
2026-02-17T08:25:01.901663+00:00 UbuntuServerPFFirewall CRON[1210]: pam_unix(cron:session): session closed for user root  
2026-02-17T08:28:10.989770+00:00 UbuntuServerPFFirewall sudo: ubuntu : TTY=ttty ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/tail -f /var/log/auth.log  
2026-02-17T08:28:10.997412+00:00 UbuntuServerPFFirewall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)  
2026-02-17T08:28:13.036429+00:00 UbuntuServerPFFirewall sudo: pam_unix(sudo:session): session closed for user root  
2026-02-17T08:28:17.975431+00:00 UbuntuServerPFFirewall sudo: ubuntu : TTY=ttty ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/tail -f /var/log/auth.log  
2026-02-17T08:28:17.975533+00:00 UbuntuServerPFFirewall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
```

Terminal Administrasi:

```
ubuntu@UbuntuServerPFFirewall:~$
```

3. Phase 2 — Attack Simulation & Detection

🎯 Tujuan Phase Ini

Menjalankan serangan brute force dari Ubuntu Desktop menggunakan Hydra, kemudian membuktikan deteksi oleh auth.log dan Fail2ban di Ubuntu Server.

3.1 Konfigurasi Fail2ban (Sebelum Serangan)

Kita set threshold lebih ketat supaya Fail2ban lebih cepat bereaksi. Jalankan di Ubuntu Server:

```
# Jalankan di: Ubuntu Server
sudo nano /etc/fail2ban/jail.local

# Isi dengan konfigurasi berikut:

[DEFAULT]
bantime = 600
findtime = 300
maxretry = 3

[sshd]
enabled = true
port = 22
filter = sshd
logpath = /var/log/auth.log
maxretry = 3
bantime = 600
```

```
# Simpan file (Ctrl+O, Enter, Ctrl+X)
# Restart fail2ban:
sudo systemctl restart fail2ban

# Verifikasi konfigurasi aktif:
sudo fail2ban-client get sshd maxretry
# Expected output: 3
```

```
ubuntu@UbuntuServerPFFirewall:~$ sudo systemctl restart fail2ban
ubuntu@UbuntuServerPFFirewall:~$ sudo fail2ban-client get sshd
/etc/fail2ban/jail.local fail2ban nano
ubuntu@UbuntuServerPFFirewall:~$ sudo fail2ban-client get sshd maxretry
3
ubuntu@UbuntuServerPFFirewall:~$
```

3.2 Simulasi Brute Force dengan Hydra

Context

Hydra adalah tool password brute-force yang digunakan oleh penetration tester.

Di sini kita gunakan dalam lab terisolasi untuk membuktikan deteksi Fail2ban.

Rockyou.txt adalah wordlist berisi password umum yang sering digunakan attacker.

Pindah ke Ubuntu Desktop — Jalankan Serangan Hydra

```
# Jalankan di: Ubuntu Desktop (Attacker)
hydra -l ubuntu -P /usr/share/wordlists/rockyou.txt \
    192.168.10.100 ssh -t 4 -V -f

# Penjelasan flag:
# -l ubuntu      : target username
# -P rockyou.txt : wordlist password
# -t 4           : 4 thread paralel
# -V             : verbose - tampilkan setiap percobaan
# -f            : stop setelah berhasil menemukan password
```

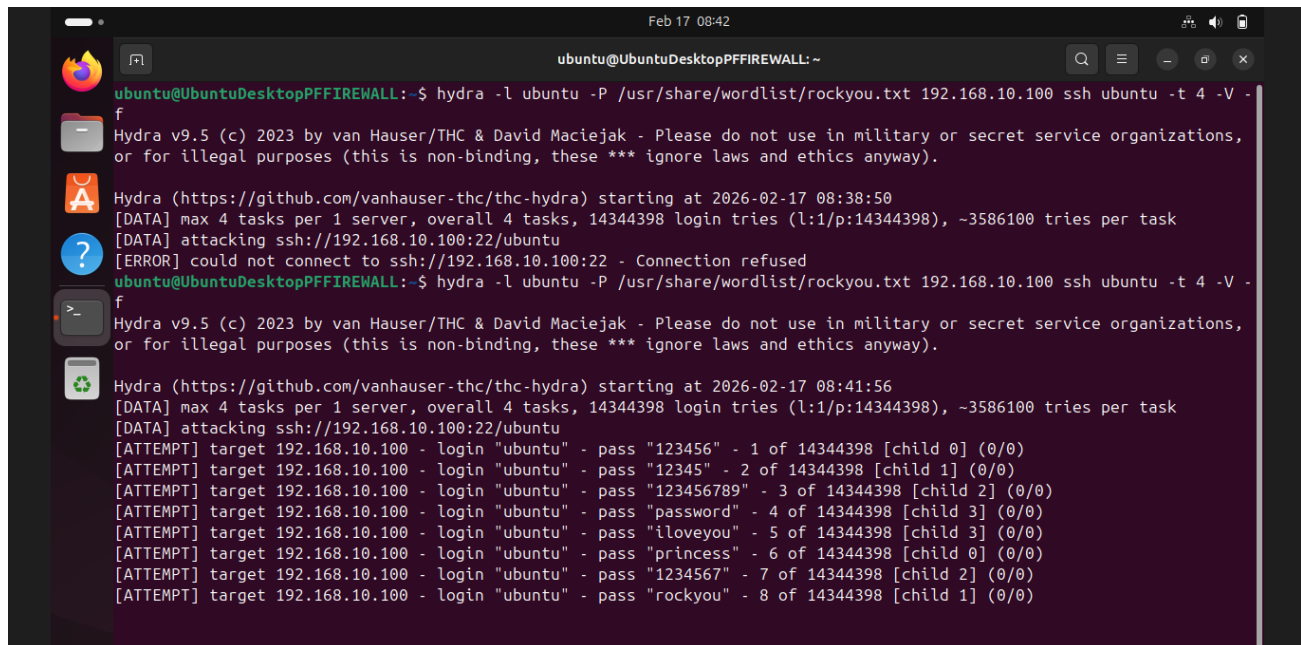
Yang Akan Terjadi

Hydra akan mencoba ratusan password dari rockyou.txt satu per satu.

Setelah 3 percobaan gagal, Fail2ban akan mendeteksi dan memblokir IP Desktop.

Kamu akan melihat pesan 'Connection refused' atau timeout dari Hydra.

Cek terminal monitoring auth.log untuk melihat log Failed password secara real-time!



```
Feb 17 08:42
ubuntu@UbuntuDesktopPFFIREWALL: ~
ubuntu@UbuntuDesktopPFFIREWALL:~$ hydra -l ubuntu -P /usr/share/wordlist/rockyou.txt 192.168.10.100 ssh ubuntu -t 4 -V -f
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations,
or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-17 08:38:50
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344398 login tries (l:1/p:14344398), ~3586100 tries per task
[DATA] attacking ssh://192.168.10.100:22/ubuntu
[ERROR] could not connect to ssh://192.168.10.100:22 - Connection refused
ubuntu@UbuntuDesktopPFFIREWALL:~$ hydra -l ubuntu -P /usr/share/wordlist/rockyou.txt 192.168.10.100 ssh ubuntu -t 4 -V -f
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations,
or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-17 08:41:56
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344398 login tries (l:1/p:14344398), ~3586100 tries per task
[DATA] attacking ssh://192.168.10.100:22/ubuntu
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "123456" - 1 of 14344398 [child 0] (0/0)
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "12345" - 2 of 14344398 [child 1] (0/0)
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "123456789" - 3 of 14344398 [child 2] (0/0)
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "password" - 4 of 14344398 [child 3] (0/0)
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "iloveyou" - 5 of 14344398 [child 3] (0/0)
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "princess" - 6 of 14344398 [child 0] (0/0)
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "1234567" - 7 of 14344398 [child 2] (0/0)
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "rockyou" - 8 of 14344398 [child 1] (0/0)
```

3.3 Observasi Detection di Ubuntu Server

Cek auth.log — Lihat Log Serangan Real-time

Di terminal monitoring Ubuntu Server, kamu akan melihat baris seperti ini:

```
ubuntu@UbuntuServerPFFirewall:~$ sudo tail -f /var/log/auth.log
2026-02-17T08:41:38.011598+00:00 UbuntuServerPFFirewall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
2026-02-17T08:41:38.059451+00:00 UbuntuServerPFFirewall sshd[1658]: Server listening on 0.0.0.0 port 22.
2026-02-17T08:41:38.059874+00:00 UbuntuServerPFFirewall sshd[1658]: Server listening on :: port 22.
2026-02-17T08:41:38.073048+00:00 UbuntuServerPFFirewall sudo: pam_unix(sudo:session): session closed for user root
2026-02-17T08:41:39.301330+00:00 UbuntuServerPFFirewall sudo: ubuntu : TTY=ttty2 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/systemctl status ssh
2026-02-17T08:41:39.302405+00:00 UbuntuServerPFFirewall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
2026-02-17T08:41:39.322897+00:00 UbuntuServerPFFirewall sudo: pam_unix(sudo:session): session closed for user root
2026-02-17T08:41:48.383728+00:00 UbuntuServerPFFirewall sudo: pam_unix(sudo:session): session closed for user root
2026-02-17T08:41:52.040732+00:00 UbuntuServerPFFirewall sudo: ubuntu : TTY=ttty1 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/tail -f /var/log/auth.log
2026-02-17T08:41:52.042258+00:00 UbuntuServerPFFirewall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
2026-02-17T08:41:58.173633+00:00 UbuntuServerPFFirewall sshd[1679]: Received disconnect from 192.168.10.101 port 40342:11: Bye Bye [preauth]
2026-02-17T08:41:58.174174+00:00 UbuntuServerPFFirewall sshd[1679]: Disconnected from authenticating user ubuntu 192.168.10.101 port 40342 [preauth]
2026-02-17T08:41:59.129182+00:00 UbuntuServerPFFirewall sshd[1684]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.10.101 user=ubuntu
2026-02-17T08:41:59.135734+00:00 UbuntuServerPFFirewall sshd[1681]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.10.101 user=ubuntu
2026-02-17T08:41:59.137296+00:00 UbuntuServerPFFirewall sshd[1682]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.10.101 user=ubuntu
2026-02-17T08:41:59.252004+00:00 UbuntuServerPFFirewall sshd[1683]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.10.101 user=ubuntu
2026-02-17T08:42:00.811790+00:00 UbuntuServerPFFirewall sshd[1684]: Failed password for ubuntu from 192.168.10.101 port 40352 ssh2
2026-02-17T08:42:00.821881+00:00 UbuntuServerPFFirewall sshd[1681]: Failed password for ubuntu from 192.168.10.101 port 40350 ssh2
2026-02-17T08:42:00.827155+00:00 UbuntuServerPFFirewall sshd[1682]: Failed password for ubuntu from 192.168.10.101 port 40348 ssh2
2026-02-17T08:42:00.956726+00:00 UbuntuServerPFFirewall sshd[1683]: Failed password for ubuntu from 192.168.10.101 port 40354 ssh2
2026-02-17T08:42:03.048892+00:00 UbuntuServerPFFirewall sshd[1684]: Failed password for ubuntu from 192.168.10.101 port 40352 ssh2
2026-02-17T08:42:03.075321+00:00 UbuntuServerPFFirewall sshd[1682]: Failed password for ubuntu from 192.168.10.101 port 40348 ssh2
2026-02-17T08:42:03.081952+00:00 UbuntuServerPFFirewall sshd[1681]: Failed password for ubuntu from 192.168.10.101 port 40350 ssh2
2026-02-17T08:42:03.475179+00:00 UbuntuServerPFFirewall sshd[1683]: Failed password for ubuntu from 192.168.10.101 port 40354 ssh2
```

Cek fail2ban.log — Konfirmasi Deteksi dan Ban

```
# Jalankan di: Ubuntu Server (Terminal A)
sudo tail -f /var/log/fail2ban.log
```

```
ubuntu@UbuntuServerPFFirewall:~$ sudo tail -f /var/log/fail2ban.log
2026-02-17 08:34:59,071 fail2ban.jail [1635]: INFO Jail 'sshd' started
2026-02-17 08:42:01,118 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2026-02-17 08:42:00
2026-02-17 08:42:01,121 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2026-02-17 08:42:00
2026-02-17 08:42:01,121 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2026-02-17 08:42:00
2026-02-17 08:42:01,122 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2026-02-17 08:42:00
2026-02-17 08:42:01,627 fail2ban.actions [1635]: NOTICE [sshd] Ban 192.168.10.101
2026-02-17 08:42:03,336 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2026-02-17 08:42:03
2026-02-17 08:42:03,338 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2026-02-17 08:42:03
2026-02-17 08:42:03,339 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2026-02-17 08:42:03
2026-02-17 08:42:03,838 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2026-02-17 08:42:03
```

Konfirmasi Status Jail SSH Setelah Serangan

```
# Jalankan di: Ubuntu Server
sudo fail2ban-client status sshd
```

```
ubuntu@UbuntuServerPFFirewall:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 1
| |- Total failed: 8
| \- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd
- Actions
| |- Currently banned: 1
| |- Total banned: 1
| \- Banned IP list: 192.168.10.101
ubuntu@UbuntuServerPFFirewall:~$
```


4. Phase 3 — Containment

🎯 Tujuan Phase Ini

Memverifikasi bahwa IP attacker benar-benar tidak bisa terhubung setelah diblokir.
Ini adalah bukti bahwa containment berhasil dilakukan secara otomatis.

4.1 Verifikasi Koneksi Ditolak dari Attacker

Kembali ke Ubuntu Desktop, coba SSH manual ke server:

```
# Jalankan di: Ubuntu Desktop (Attacker)
ssh ubuntu@192.168.10.100

# Expected: Connection timed out ATAU Connection refused
# IP 192.168.10.101 sudah diblokir oleh Fail2ban — tidak bisa konek!

ubuntu@UbuntuDesktopPFFIREWALL:~$ ssh ubuntu@192.168.10.100
ssh: connect to host 192.168.10.100 port 22: Connection refused

# Coba juga ping untuk konfirmasi network masih jalan tapi SSH ditolak:
ping 192.168.10.100 -c 3
# Ping harus tetap reply (ICMP tidak diblokir, hanya SSH port 22)

ubuntu@UbuntuDesktopPFFIREWALL:~$ ping 192.168.10.100
PING 192.168.10.100 (192.168.10.100) 56(84) bytes of data.
64 bytes from 192.168.10.100: icmp_seq=23 ttl=64 time=2.43 ms
64 bytes from 192.168.10.100: icmp_seq=24 ttl=64 time=3.25 ms
64 bytes from 192.168.10.100: icmp_seq=25 ttl=64 time=2.41 ms
64 bytes from 192.168.10.100: icmp_seq=26 ttl=64 time=1.45 ms
64 bytes from 192.168.10.100: icmp_seq=27 ttl=64 time=320 ms
64 bytes from 192.168.10.100: icmp_seq=28 ttl=64 time=1.27 ms
64 bytes from 192.168.10.100: icmp_seq=29 ttl=64 time=2.01 ms
^C
--- 192.168.10.100 ping statistics ---
29 packets transmitted, 7 received, 75.8621% packet loss, time 28721ms
rtt min/avg/max/mdev = 1.267/47.553/320.054/111.249 ms
```

5. Phase 4 — Eradication

🎯 Tujuan Phase Ini

Memastikan tidak ada akses berhasil (zero successful login) selama serangan berlangsung.
Verifikasi integritas sistem — tidak ada user baru atau backdoor yang dibuat attacker.

5.1 Audit Login History

```
# Jalankan di: Ubuntu Server
```

```
# Cek login yang BERHASIL — harus 0 dari 192.168.10.101 saat serangan  
sudo grep -a 'Accepted password' /var/log/auth.log
```

```
ubuntu@UbuntuServerPFFirewall:~$ sudo grep -a 'Accepted password' /var/log/auth.log  
[sudo] password for ubuntu:  
2026-02-16T09:12:41.139750+00:00 UbuntuServerPFFirewall sshd[1775]: Accepted password for ubuntu from 192.168.10.101 port 46882 ssh2  
2026-02-16T09:19:49.247343+00:00 UbuntuServerPFFirewall sshd[1857]: Accepted password for ubuntu from 192.168.10.101 port 56558 ssh2  
2026-02-16T09:26:33.699170+00:00 UbuntuServerPFFirewall sshd[1974]: Accepted password for ubuntu from 192.168.10.101 port 36016 ssh2  
2026-02-16T09:33:39.450471+00:00 UbuntuServerPFFirewall sshd[2243]: Accepted password for ubuntu from 192.168.10.101 port 55908 ssh2  
2026-02-16T12:20:09.212102+00:00 UbuntuServerPFFirewall sshd[975]: Accepted password for ubuntu from 192.168.10.101 port 35850 ssh2  
2026-02-17T15:10:55.776116+00:00 UbuntuServerPFFirewall sshd[1004]: Accepted password for ubuntu from 192.168.10.101 port 39360 ssh2  
2026-02-17T15:24:50.527575+00:00 UbuntuServerPFFirewall sudo: ubuntu : TTY=ttty1 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/grep 'Accepted password' /var/log/auth.log  
2026-02-17T15:33:30.570459+00:00 UbuntuServerPFFirewall sshd[1308]: Accepted password for ubuntu from 192.168.10.101 port 49386 ssh2  
2026-02-17T16:14:06.371589+00:00 UbuntuServerPFFirewall sudo: ubuntu : TTY=ttty1 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/grep -a 'Accepted password' /var/log/auth.log  
ubuntu@UbuntuServerPFFirewall:~$ _
```

Output di atas menunjukkan sesi SSH legitimate dari admin (Feb 16 dan Feb 17 15:13-15:33) yang dilakukan sebelum dan sesudah lab setup. Tidak ada 'Accepted password' yang tercatat pada window serangan Hydra yaitu 08:41-08:42, membuktikan zero successful login selama brute force berlangsung."

```
sudo grep -a 'Accepted publickey' /var/log/auth.log
```

```
ubuntu@UbuntuServerPFFirewall:~$ sudo grep -a 'Accepted publickey' /var/log/auth.log  
2026-02-17T15:24:43.224542+00:00 UbuntuServerPFFirewall sudo: ubuntu : TTY=ttty1 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/grep 'Accepted publickey' /var/log/auth.log  
2026-02-17T15:24:53.920437+00:00 UbuntuServerPFFirewall sudo: ubuntu : TTY=ttty1 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/grep 'Accepted publickey' /var/log/auth.log  
2026-02-17T15:14:49.626294+00:00 UbuntuServerPFFirewall sudo: ubuntu : TTY=ttty1 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/grep -a 'Accepted publickey' /var/log/auth.log  
ubuntu@UbuntuServerPFFirewall:~$ _
```

```
# Lihat riwayat login lengkap:  
sudo last | head -20
```

```
ubuntu@UbuntuServerPFFirewall:~$ sudo last | head -20
ubuntu pts/0 192.168.10.101 Tue Feb 17 15:33 - 15:33 (00:00)
ubuntu pts/0 192.168.10.101 Tue Feb 17 15:13 - 15:21 (00:07)
reboot system boot 6.8.0-100-generi Tue Feb 17 15:14 still running
reboot system boot 6.8.0-100-generi Tue Feb 17 08:16 still running
ubuntu pts/0 192.168.10.101 Mon Feb 16 12:20 - crash (19:56)
reboot system boot 6.8.0-100-generi Mon Feb 16 12:15 still running
ubuntu pts/2 192.168.10.101 Mon Feb 16 09:33 - crash (02:41)
ubuntu pts/2 192.168.10.101 Mon Feb 16 09:26 - 09:33 (00:06)
ubuntu pts/1 192.168.10.101 Mon Feb 16 09:19 - crash (02:55)
ubuntu pts/0 192.168.10.101 Mon Feb 16 09:12 - crash (03:02)
reboot system boot 6.8.0-100-generi Mon Feb 16 09:02 still running
reboot system boot 6.8.0-100-generi Mon Feb 16 09:02 still running
reboot system boot 6.8.0-100-generi Mon Feb 16 09:00 still running
reboot system boot 6.8.0-100-generi Mon Feb 16 08:55 still running
reboot system boot 6.8.0-100-generi Mon Feb 16 08:54 still running
reboot system boot 6.8.0-100-generi Mon Feb 16 08:51 still running
reboot system boot 6.8.0-100-generi Mon Feb 16 08:49 still running
reboot system boot 6.8.0-100-generi Mon Feb 16 08:48 still running
reboot system boot 6.8.0-100-generi Mon Feb 16 08:47 still running
reboot system boot 6.8.0-100-generi Mon Feb 16 08:42 still running
ubuntu@UbuntuServerPFFirewall:~$ _
```

Cek user yang sedang login saat ini:

who

w

```
ubuntu@UbuntuServerPFFirewall:~$ who
ubuntu tty1 2026-02-17 15:14
ubuntu@UbuntuServerPFFirewall:~$ w
 16:16:05 up 1:03, 1 user, load average: 0.14, 0.18, 0.17
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU WHAT
ubuntu    tty1      -                15:14    5.00s  0.59s  ?    w
ubuntu@UbuntuServerPFFirewall:~$ _
```

5.2 Verifikasi Integritas User

Jalankan di: Ubuntu Server

Pastikan tidak ada user baru yang dibuat attacker

sudo cat /etc/passwd | tail -5

```
ubuntu@UbuntuServerPFFirewall:~$ sudo cat /etc/passwd | tail -5
landscape:x:107:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
ubuntu:x:1000:1000:ubuntu:/home/ubuntu:/bin/bash
sshd:x:109:65534::/run/ssh:/usr/sbin/nologin
ubuntu@UbuntuServerPFFirewall:~$
```

Cek sudo users

sudo cat /etc/sudoers.d/*

sudo getent group sudo

```
ubuntu@UbuntuServerPFFirewall:~$ sudo cat /etc/sudoers.d/*
#
# The default /etc/sudoers file created on installation of the
# sudo package now includes the directive:
#
#     @includedir /etc/sudoers.d
#
# This will cause sudo to read and parse any files in the /etc/sudoers.d
# directory that do not end in '~' or contain a '.' character, if it
# exists. It is not an error if the directory does not exist.
#
# Note also, that because sudoers contents can vary widely, no attempt is
# made to add this directive to existing sudoers files on upgrade. Feel free
# to add the above directive to the end of your /etc/sudoers file to enable
# this functionality for existing installations if you wish!
# Sudo versions older than 1.9.1 will only support the old syntax
# #includedir. That means that the sudo versions in Debian bullseye (11)
# and later will happily accept both @includedir and #includedir.
#
# Finally, please note that using the visudo command is the recommended way
# to update sudoers content, since it protects against many failure modes.
# See the man page for visudo and sudoers for more information.
#
ubuntu@UbuntuServerPFFirewall:~$ sudo getent group sudo
sudo:x:27:ubuntu
ubuntu@UbuntuServerPFFirewall:~$
```

```
# Cek proses mencurigakan
ps aux | grep -v '^[a-z]' | head -20
```

```
ubuntu@UbuntuServerPFFirewall:~$ ps aux | grep -v '^[a-z]' | head -20
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
ubuntu@UbuntuServerPFFirewall:~$ _
```

6. Phase 5 — Recovery

🎯 Tujuan Phase Ini

Melakukan unban IP untuk membuktikan sistem bisa recovery secara terkontrol.
Verifikasi bahwa akses legitimate dari Desktop kembali berfungsi normal setelah unban.

6.1 Manual Unban IP Attacker

Dalam skenario recovery, admin bisa unban IP legitimate yang salah terblokir, atau menunggu bantime (600 detik) habis secara otomatis. Di lab ini kita unban secara manual:

```
# Jalankan di: Ubuntu Server
sudo fail2ban-client set sshd unbanip 192.168.10.101

# Verifikasi unban berhasil:
sudo fail2ban-client status sshd
# Expected: Currently banned: 0, Banned IP list: (kosong)
```

```
ubuntu@UbuntuServerPFFirewall:~$ sudo fail2ban-client set sshd unbanip 192.168.10.101
0
ubuntu@UbuntuServerPFFirewall:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:    0
|   \- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd
|- Actions
|   |- Currently banned: 0
|   |- Total banned:    0
|   \- Banned IP list:
ubuntu@UbuntuServerPFFirewall:~$ _
```

6.2 Verifikasi Akses Legitimate Kembali Normal

Kembali ke Ubuntu Desktop, coba SSH kembali dengan kredensial yang benar:

```
# Jalankan di: Ubuntu Desktop
ssh ubuntu@192.168.10.100

# Masukkan password YANG BENAR saat diminta
# Expected: Login berhasil setelah unban

# Setelah masuk, verifikasi:
whoami
hostname
exit
```

```
Last login: Tue Feb 17 15:13:58 2026 from 192.168.10.101
ubuntu@UbuntuServerPFFirewall:~$ whoami
ubuntu
ubuntu@UbuntuServerPFFirewall:~$ hostname
UbuntuServerPFFirewall
ubuntu@UbuntuServerPFFirewall:~$ exit
logout
Connection to 192.168.10.100 closed.
ubuntu@UbuntuDesktopPFFIREWALL:~$
```

6.3 Final System Check

```
# Jalankan di: Ubuntu Server
```

```
# Pastikan semua service masih berjalan normal:
sudo systemctl status ssh
```

```
ubuntu@UbuntuServerPFFirewall:~$ sudo systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-02-17 15:34:42 UTC; 15s ago
     TriggeredBy: • ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Main PID: 1397 (sshd)
      Tasks: 1 (limit: 4606)
     Memory: 1.2M (peak: 1.5M)
        CPU: 35ms
    CGroup: /system.slice/ssh.service
            └─1397 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 17 15:34:42 UbuntuServerPFFirewall systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 17 15:34:42 UbuntuServerPFFirewall sshd[1397]: Server listening on 0.0.0.0 port 22.
Feb 17 15:34:42 UbuntuServerPFFirewall sshd[1397]: Server listening on :: port 22.
Feb 17 15:34:42 UbuntuServerPFFirewall systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
ubuntu@UbuntuServerPFFirewall:~$ _
```

```
sudo systemctl status fail2ban
```

```
ubuntu@UbuntuServerPFFirewall:~$ sudo systemctl status fail2ban
• fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-02-17 15:14:13 UTC; 21min ago
     Docs: man:fail2ban(1)
    Main PID: 694 (fail2ban-server)
      Tasks: 5 (limit: 4606)
     Memory: 37.5M (peak: 37.8M)
        CPU: 13.964s
    CGroup: /system.slice/fail2ban.service
            └─694 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Feb 17 15:14:13 UbuntuServerPFFirewall systemd[1]: Started fail2ban.service - Fail2Ban Service.
Feb 17 15:14:20 UbuntuServerPFFirewall fail2ban-server[694]: 2026-02-17 15:14:20,321 fail2ban.configreader
Feb 17 15:14:25 UbuntuServerPFFirewall fail2ban-server[694]: Server ready
lines 1-14/14 (END)
```

```
sudo ufw status verbose
```

```
ubuntu@UbuntuServerPFFirewall:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
```

To	Action	From
--	-----	----
22/tcp (OpenSSH)	ALLOW IN	Anywhere
80/tcp	ALLOW IN	Anywhere
22/tcp (OpenSSH (v6))	ALLOW IN	Anywhere (v6)
80/tcp (v6)	ALLOW IN	Anywhere (v6)

```
ubuntu@UbuntuServerPFFirewall:~$ _
```

```
# Summary log insiden:
```

```
sudo grep '192.168.10.101' /var/log/fail2ban.log
```

```
ubuntu@UbuntuServerPFFirewall:~$ sudo grep '192.168.10.101' /var/log/fail2ban.log
2025-02-17 08:42:01,118 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2025-02-17 08:42:00
2025-02-17 08:42:01,121 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2025-02-17 08:42:00
2025-02-17 08:42:01,121 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2025-02-17 08:42:00
2025-02-17 08:42:01,122 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2025-02-17 08:42:00
2025-02-17 08:42:01,627 fail2ban.actions [1635]: NOTICE [sshd] Ban 192.168.10.101
2025-02-17 08:42:03,336 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2025-02-17 08:42:03
2025-02-17 08:42:03,338 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2025-02-17 08:42:03
2025-02-17 08:42:03,339 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2025-02-17 08:42:03
2025-02-17 08:42:03,833 fail2ban.filter [1635]: INFO [sshd] Found 192.168.10.101 - 2025-02-17 08:42:03
2025-02-17 08:52:03,309 fail2ban.actions [1635]: NOTICE [sshd] Unban 192.168.10.101
ubuntu@UbuntuServerPFFirewall:~$ _
```

7. Incident Response Documentation

Incident Report Card

Incident ID	IR-2026-001
Tanggal	17/02/2026
Severity	Medium
Jenis Serangan	SSH Brute Force Attack
Attacker IP	192.168.10.101 (Ubuntu Desktop)
Target IP	192.168.10.100 (Ubuntu Server)
Tool Digunakan	Hydra (wordlist: rockyou.txt)
Status Akhir	Contained — Tidak ada akses berhasil
Ditangani Oleh	GLENVIO REGALITO RAHARDJO

8. Lessons Learned & Rekomendasi

8.1 Apa yang Berhasil

- ✓ Fail2ban berhasil mendeteksi dan memblokir serangan brute force SSH secara otomatis dalam hitungan detik
- ✓ auth.log memberikan visibility real-time terhadap percobaan login yang gagal
- ✓ UFW firewall membatasi attack surface — hanya port 22 dan 80 yang terbuka
- ✓ Tidak ada akses berhasil selama serangan berlangsung — sistem tetap aman
- ✓ Kombinasi UFW + SSH Hardening + Fail2ban terbukti efektif sebagai Defense in Depth

8.2 Indicator of Compromise (IoC)

IoC Type	Indicator	Source Log
Network IoC	IP 192.168.10.101 — Multiple failed SSH	<code>/var/log/auth.log</code>
Behavioral IoC	≥3 Failed password dalam 300 detik	<code>/var/log/auth.log</code>
Tool Signature	Rapid SSH connection attempts (Hydra)	<code>/var/log/auth.log</code>
Response	Auto-ban triggered by Fail2ban	<code>/var/log/fail2ban.log</code>

8.3 Rekomendasi Peningkatan

Short-term (Bisa dilakukan sekarang)

- SSH Key Authentication:** Disable password auth, gunakan hanya public key — eliminasi 100% brute force risk
- Change SSH Port:** Ganti port SSH dari 22 ke port custom (misal 2222) untuk mengurangi noise dari script scanner
- Increase bantime:** Set bantime = 86400 (24 jam) atau -1 (permanent) untuk attacker yang lebih agresif

Long-term (Next Project)

- SIEM Integration:** Wazuh atau Splunk untuk centralized log monitoring dan alerting
- IDS/IPS:** Snort atau Suricata untuk signature-based attack detection di level network
- Automated Alerting:** Konfigurasi Fail2ban untuk kirim email/notifikasi saat ada IP yang diblokir

9. Conclusion

Project Brute Force Attack Simulation dan Incident Response ini berhasil membuktikan secara praktis bahwa implementasi Defense in Depth pada Ubuntu Server mampu mendeteksi dan menahan serangan brute force SSH secara otomatis tanpa intervensi manual.

Melalui simulasi menggunakan Hydra dari attacker node, sistem berhasil:

- **Mendeteksi** serangan melalui auth.log dalam hitungan detik
- **Memblokir** IP attacker secara otomatis melalui Fail2ban setelah maxretry terpenuhi
- **Mempertahankan** zero successful login selama serangan berlangsung
- **Recover** dengan bersih melalui proses unban terkontrol

Key Takeaway

Defense in Depth bukan hanya tentang blocking — tapi tentang layered visibility.
Kombinasi UFW (perimeter) + SSH Hardening (access) + Fail2ban (automated response)
menciptakan sistem yang tidak hanya aman, tapi juga bisa merespons ancaman secara mandiri.

Ini adalah fondasi SOC (Security Operations Center) yang sesungguhnya.

Ethical & Legal Reminder

Seluruh aktivitas dalam modul ini dilakukan pada sistem milik sendiri dalam lab terisolasi.
Brute force attack terhadap sistem tanpa izin adalah tindakan ilegal.
Gunakan pengetahuan ini hanya untuk defensive security dan ethical hacking yang terotorisasi.